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ShowName: Cisco CCNA (200-301)

Topic: 1.0 Networking Fundamentals

EpisodeName: Networking Topology Architectures: WAN

Description: In this episode, Vonne and Ronnie introduce the different WAN access topology architectures that network technicians need to be able to identify and understand. The team describe the basic topologies implemented regardless of whether you're using MPLS, MetroE, Broadband PPPoE or VPNs for connectivity

Networking Topology Architectures: WAN

1.2 Describe characteristics of network topology architectures

1.2.d WAN

Topologies

- Point-to-Point
- Hub-and-Spoke
- Full Mesh
- Single/Dual-homed

WAN Access types

MPLS (Multi Protocol Label Switching)

Purpose:

- Not dependent on any single layer 2 technology
- Carry any payload: IPv4, IPv6, Ethernet, ATM, DSL, FR

ISP WAN technology

Metro Ethernet (MetroE)

Purpose:

- Ethernet based connectivity to internet or other WAN
- Provides for ethernet speeds to multiple sites within same geography
- Cheaper than SONET

MAN and connection to Internet

Broadband PPPoE

Purpose:

- high speed connections and always on (compared to dial-up)
- to add authentication and encryption to always on network

Cable internet access etc.

Internet VPNs

- Site-to-Site
- Client Access VPN
- DMVPN (Dynamic Multipoint VPN)